Circles, Radians, Trig & Conditional Probability

ClipSAT Printable Practice Suite (clipsat.org) • Topic 10.01 • 20 FRQs + 25 MCQs • Difficulty Progression • Cross-Lesson Synthesis

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • [FRQ] • Target: Standard Equation of a Circle

Find the center (h, k) and radius r of the circle $(x-5)^2+(y+3)^2=49$, and state the coordinates of the four points on the circle that lie horizontally or vertically from the center.

Comparing with $(x-h)^2+(y-k)^2=r^2$, the center is $(5,\ -3)$ and the radius is $r=\sqrt{49}=7$. Moving ± 7 horizontally and vertically gives $(12,\ -3), (-2,\ -3), (5, 4), (5,\ -10)$.

2 [Foundational] • [FRQ] • Target: Completing the Square for Circle Equations

Convert the general circle equation $x^2+y^2-8x+6y-11=0$ into standard form by completing the square, and find its center and radius.

Group and complete the square: $(x^2-8x+16)+(y^2+6y+9)=11+16+9=36 \Rightarrow (x-4)^2+(y+3)^2=36$. The center is $(4,\ -3)$ and the radius is $r=6$.

3 [Foundational] • [FRQ] • Target: Degree-Radian Conversions

Convert (a) 135° and 210° to radians, and (b) $\frac{5\pi}{6}$ and $\frac{7\pi}{4}$ radians to degrees.

(a) $135^\circ \times \frac{\pi}{180^\circ}=\frac{3\pi}{4}$ rad; $210^\circ \times \frac{\pi}{180^\circ}=\frac{7\pi}{6}$ rad. (b) $\frac{5\pi}{6}\times\frac{180^\circ}{\pi}=150^\circ$; $\frac{7\pi}{4}\times\frac{180^\circ}{\pi}=315^\circ$.

4 [Foundational] • [FRQ] • Target: Arc Length & Sector Area

In a circle of radius $r=12$ cm shown below, a central angle measures $\theta=60^\circ$ ($\frac{\pi}{3}$ rad). Find the exact arc length s and the exact area A of the shaded sector in terms of π.

Arc length $s=r\theta=12\left(\frac{\pi}{3}\right)=4\pi$ cm. Sector area $A=\frac{1}{2}r^2\theta=\frac{1}{2}(144)\left(\frac{\pi}{3}\right)=24\pi$ cm^2.

5 [Foundational] • [FRQ] • Target: Inscribed Angle Theorem

On a circle with center O, an inscribed angle $\angle ABC$ intercepts minor arc AC. If $\angle ABC=38^\circ$, find the degree measure of minor arc AC and central angle $\angle AOC$.

By the Inscribed Angle Theorem, the intercepted arc and central angle are twice the inscribed angle: $m\widehat{AC}=\angle AOC=2(38^\circ)=76^\circ$.

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6 [Foundational] • [FRQ] • Target: Unit Circle Trigonometry

Evaluate the exact values of (a) $\sin\left(\frac{2\pi}{3}\right)$, (b) $\cos\left(\frac{5\pi}{4}\right)$, and (c) $\tan\left(\frac{11\pi}{6}\right)$ using reference angles on the unit circle.

(a) $\frac{2\pi}{3}$ is in QII with reference angle $\frac{\pi}{3}$, so $\sin\left(\frac{2\pi}{3}\right) = \frac{\sqrt{3}}{2}$. (b) $\frac{5\pi}{4}$ is in QIII with reference angle $\frac{\pi}{4}$, so $\cos\left(\frac{5\pi}{4}\right) = -\frac{\sqrt{2}}{2}$. (c) $\frac{11\pi}{6}$ is in QIV with reference angle $\frac{\pi}{6}$, so $\tan\left(\frac{11\pi}{6}\right) = -\frac{\sqrt{3}}{3}$.

7 [Foundational] • [FRQ] • Target: Tangent-Radius Perpendicularity

Point P is located $17\,\text{cm}$ from the center O of a circle of radius $8\,\text{cm}$. A line through P is tangent to the circle at point T. Find the length of the tangent segment PT.

Radius OT is perpendicular to tangent PT at T ($\angle OTP = 90^\circ$). By the Pythagorean theorem, $PT = \sqrt{OP^2 - OT^2} = \sqrt{17^2 - 8^2} = \sqrt{225} = 15\,\text{cm}$.

Part I — Free-Response Questions (FRQs): Intermediate Modeling (8–14)

8 [Intermediate] • [FRQ] • Target: Chord Bisector & Distance from Center

In a circle of radius $r = 10\,\text{cm}$, a chord AB has length $16\,\text{cm}$, as shown. Find the perpendicular distance d from the center O to chord AB.

The perpendicular from the center O to chord AB bisects the chord into two segments of length $8\,\text{cm}$. In the resulting right triangle with hypotenuse $r = 10\,\text{cm}$ and leg $8\,\text{cm}$, $d = \sqrt{10^2 - 8^2} = 6\,\text{cm}$.

9 [Intermediate] • [FRQ] • Target: Equation of a Tangent Line to a Circle

Find the equation of the tangent line to the circle $(x-1)^2 + (y+2)^2 = 25$ at the point $T(4, 2)$ on the circle.

The center is $C(1, -2)$. The slope of radius CT is $m_{CT} = \frac{2-(-2)}{4-1} = \frac{4}{3}$. Therefore, the tangent line at $T(4, 2)$ has perpendicular slope $m_\perp = -\frac{3}{4}$ and equation $y - 2 = -\frac{3}{4}(x-4) \Rightarrow y = -\frac{3}{4}x + 5$.

10 [Intermediate] • [FRQ] • Target: Pythagorean Identity & Quadrant Signs

If $\cos\theta = -\frac{5}{13}$ and θ lies in Quadrant III ($\pi < \theta < \frac{3\pi}{2}$), find the exact values of $\sin\theta$ and $\tan\theta$.

In Quadrant III, $\sin\theta < 0$ and $\tan\theta > 0$. By $\sin^2\theta + \cos^2\theta = 1$, $\sin\theta = -\sqrt{1 - (-5/13)^2} = -\frac{12}{13}$, and $\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{12}{5}$.

11 [Intermediate] • [FRQ] • Target: Relating Sector Area & Arc Length ($A = \frac{1}{2}rs$)

A circular sector has arc length $6\pi\,\text{cm}$ and area $27\pi\,\text{cm}^2$. Find the radius r of the circle and the central angle θ in both radians and degrees.

Divide $A = \frac{1}{2}rs$ by s: $27\pi = \frac{1}{2}r(6\pi) \Rightarrow 3\pi r = 27\pi \Rightarrow r = 9\,\text{cm}$. Then $\theta = \frac{s}{r} = \frac{6\pi}{9} = \frac{2\pi}{3}\,\text{rad} = 120^\circ$.

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12 [Intermediate] • [FRQ] • Target: Non-Monic Circle Equations

Find the diameter and area of the circle whose equation is $2x^2 + 2y^2 + 12x - 8y - 24 = 0$.

Divide by 2: $x^2 + y^2 + 6x - 4y - 12 = 0$. Complete the square: $(x+3)^2 + (y-2)^2 = 12 + 9 + 4 = 25$. Thus $r = 5$, diameter $= 2r = 10$, and area $= \pi r^2 = 25\pi$.

13 [Intermediate] • [FRQ] • Target: Cyclic Quadrilaterals

Quadrilateral $ABCD$ is inscribed in a circle (cyclic quadrilateral). If $\angle A = (2x + 15)^\circ$ and $\angle C = (3x - 10)^\circ$, find x, $\angle A$, and $\angle C$.

Opposite angles of a cyclic quadrilateral are supplementary:

$(2x + 15) + (3x - 10) = 180 \Rightarrow 5x + 5 = 180 \Rightarrow 5x = 175 \Rightarrow x = 35$. Thus $\angle A = 85^\circ$ and $\angle C = 95^\circ$.

14 [Intermediate] • [FRQ] • Target: Sinusoidal Function Parameters

State the amplitude, period, midline, and maximum value of the sinusoidal function $f(t) = 4\sin\left(\frac{\pi}{6}t\right) + 7$, as graphed below.

Amplitude $= |4| = 4$; period $= \frac{2\pi}{\pi/6} = 12$; midline is $y = 7$; maximum value is $7 + 4 = 11$ (and minimum value is $7 - 4 = 3$).

Part I — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • [FRQ] • Target: Area of a Circular Segment • Cross-Lesson: Topic 8.01 (Composite Areas)

Find the exact area of the circular segment (the region between chord AB and minor arc AB) in a circle of radius $r = 6\,\text{cm}$ if the central angle subtended by AB is 90°.

Sector area ($90^\circ = \frac{1}{4}$ circle) $= \frac{1}{4}\pi(6^2) = 9\pi\,\text{cm}^2$. Right triangle OAB has area $\frac{1}{2}(6)(6) = 18\,\text{cm}^2$. Circular segment area $= (9\pi - 18)\,\text{cm}^2$.

16 [Challenging] • [FRQ] • Target: Line-Circle Secant Intersection System • Cross-Lesson: Topic 3.01 (Quadratics) & Topic 2.01 (Systems)

Find the coordinates of all intersection points of the line $y = x + 1$ and the circle $x^2 + y^2 = 25$, and compute the length of the secant chord.

Substitute $y = x + 1$: $x^2 + (x+1)^2 = 25 \Rightarrow 2x^2 + 2x - 24 = 0 \Rightarrow x^2 + x - 12 = 0 \Rightarrow (x+4)(x-3) = 0$. The intersection points are $(-4, -3)$ and $(3, 4)$, and the chord length is $\sqrt{(3-(-4))^2 + (4-(-3))^2} = \sqrt{49 + 49} = 7\sqrt{2}$.

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17 [Challenging] • [FRQ] • Target: Power of a Point (Tangent-Secant Theorem) • Cross-Lesson: Topic 7.01 (Similar Triangles)

From an external point P, a tangent segment PT of length 12 and a secant line PAB (passing through the circle at A and B) are drawn. If the external secant part is $PA = 8$, find the length of the internal chord AB.

By the Power of a Point (Tangent-Secant) Theorem, $PT^2 = PA \cdot PB \Rightarrow 12^2 = 8(8 + AB) \Rightarrow 144 = 64 + 8 \cdot AB \Rightarrow 8 \cdot AB = 80 \Rightarrow AB = 10$.

18 [Challenging] • [FRQ] • Target: Periodic Trigonometric Modeling • Cross-Lesson: Topic 4.01 (Function Modeling)

A Ferris wheel has a diameter of $40\,\text{m}$, its lowest point is $4\,\text{m}$ above the ground, and it completes one full revolution every 60 seconds. If a rider boards at the lowest point at $t = 0$, write a cosine function $h(t)$ for the rider's height above the ground at time t seconds, and find $h(15)$ and $h(30)$.

Radius $A = 20\,\text{m}$, center height (midline) $D = 4 + 20 = 24\,\text{m}$, angular frequency $\omega = \frac{2\pi}{60} = \frac{\pi}{30}$. Starting at the minimum at $t = 0$: $h(t) = -20\cos\left(\frac{\pi}{30}t\right) + 24$. Thus $h(15) = 24\,\text{m}$ and $h(30) = 44\,\text{m}$.

19 [Challenging] • [FRQ] • Target: Tangent Line Parameter via Point-to-Line Distance • Cross-Lesson: Topic 1.01 (Linear Equations & Absolute Value)

For what values of c is the line $3x + 4y = c$ tangent to the circle $x^2 + y^2 - 2x + 4y = 20$?

Rewrite the circle as $(x - 1)^2 + (y + 2)^2 = 25$, which has center $(1, -2)$ and radius $r = 5$. For tangency, the perpendicular distance from $(1, -2)$ to $3x + 4y - c = 0$ must equal 5:
$\frac{|3(1) + 4(-2) - c|}{\sqrt{3^2 + 4^2}} = 5 \Rightarrow |-5 - c| = 25 \Rightarrow c + 5 = \pm 25 \Rightarrow c = 20$ or $c = -30$.

20 [Challenging] • [FRQ] • Target: Common External Tangent of Two Circles • Cross-Lesson: Topic 7.01 (Pythagorean Theorem)

Two circles of radii $r_1 = 9\,\text{cm}$ and $r_2 = 4\,\text{cm}$ are externally tangent to each other and both are tangent to the same straight line ℓ. Find the distance between the two points of tangency on line ℓ.

The centers are $r_1 + r_2 = 13\,\text{cm}$ apart, and the difference in their heights above line ℓ is $r_1 - r_2 = 5\,\text{cm}$. By the Pythagorean theorem, the horizontal distance between the points of tangency is $d = \sqrt{(r_1 + r_2)^2 - (r_1 - r_2)^2} = 2\sqrt{r_1 r_2} = 2\sqrt{36} = 12\,\text{cm}$.

Part II — Multiple-Choice Questions (MCQs): Foundational Concepts (21–28)

21 [Foundational] • [MCQ] • Target: Reading Circle Radius from Standard Form

In the xy-plane, what is the radius of the circle $(x + 6)^2 + (y - 8)^2 = 100$?

a. 6

b. 8

c. 10 ✓

d. 100

From $(x - h)^2 + (y - k)^2 = r^2 = 100$, the radius is $r = \sqrt{100} = 10$.

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22 [Foundational] • [MCQ] • Target: Radians to Degrees

An angle has a measure of $\frac{5\pi}{12}$ radians. What is the measure of the angle in degrees?

a. 60°

b. 75° ✓

c. 80°

d. 105°

$\frac{5\pi}{12} \times \frac{180^\circ}{\pi} = 5 \times 15^\circ = 75^\circ$.

23 [Foundational] • [MCQ] • Target: Arc Length Formula $s = r\theta$

In a circle with radius 8, what is the length of an arc intercepted by a central angle of $\frac{3\pi}{4}$ radians?

a. 3π

b. 6π ✓

c. 12π

d. 24π

Arc length $s = r\theta = 8 \cdot \frac{3\pi}{4} = 6\pi$.

24 [Foundational] • [MCQ] • Target: Testing Points on a Circle

Which of the following points lies on the circle $(x-2)^2 + (y+1)^2 = 25$?

a. $(2, 5)$

b. $(5, 3)$ ✓

c. $(-2, 3)$

d. $(6, 2)$

Test $(5, 3)$: $(5-2)^2 + (3+1)^2 = 3^2 + 4^2 = 9 + 16 = 25$.

25 [Foundational] • [MCQ] • Target: Sector Area as Fraction of Circle

A circle has circumference 16π. What is the area of a 45° sector of this circle?

a. 2π

b. 4π

c. 8π ✓

d. 16π

From $2\pi r = 16\pi$, we get $r = 8$ and total circle area $\pi r^2 = 64\pi$. A 45° sector is $\frac{45^\circ}{360^\circ} = \frac{1}{8}$ of the circle: $\frac{1}{8}(64\pi) = 8\pi$.

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26 [Foundational] • [MCQ] • Target: Quadrant II Cosine Sign

If $\sin\theta = \frac{4}{5}$ and θ is in Quadrant II, what is the value of $\cos\theta$?

a. $-\frac{4}{5}$

b. $-\frac{3}{5}$ ✓

c. $\frac{3}{5}$

d. $-\frac{3}{4}$

In Quadrant II, $\cos\theta < 0$, so $\cos\theta = -\sqrt{1-(4/5)^2} = -\frac{3}{5}$.

27 [Foundational] • [MCQ] • Target: Finding Center by Completing the Square

What is the center of the circle given by $x^2 + y^2 - 10x + 4y = 7$?

a. $(-5, 2)$

b. $(5, -2)$ ✓

c. $(-10, 4)$

d. $(10, -4)$

Complete the square: $(x-5)^2 + (y+2)^2 = 7 + 25 + 4 = 36$, so the center is $(5, -2)$.

28 [Foundational] • [MCQ] • Target: Inscribed Angle vs. Central Angle

Points $A, B,$ and C lie on a circle with center O. If central angle $\angle AOC = 110^\circ$ and B lies on the major arc AC, what is the measure of inscribed angle $\angle ABC$?

a. 55° ✓

b. 70°

c. 110°

d. 125°

By the Inscribed Angle Theorem, $\angle ABC = \frac{1}{2}\angle AOC = \frac{1}{2}(110^\circ) = 55^\circ$.

Part II — Multiple-Choice Questions (MCQs): Intermediate Exam Practice (29–37)

29 [Intermediate] • [MCQ] • Target: Finding Constant c from Circle Radius

The equation $x^2 + y^2 + 6x - 14y + c = 0$ represents a circle with radius 4. What is the value of the constant c?

a. 34

b. 42 ✓

c. 50

d. 74

Complete the square: $(x+3)^2 + (y-7)^2 = 9 + 49 - c = 58 - c$. Setting $r^2 = 58 - c = 4^2 = 16$ gives $c = 42$.

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30 [Intermediate] • [MCQ] • Target: Circle Equation from Diameter Endpoints

In the xy-plane, segments AB is a diameter of a circle. If $A = (-3, 2)$ and $B = (5, 8)$, what is the equation of the circle?

a. $(x-1)^2 + (y-5)^2 = 25$ ✓

b. $(x-1)^2 + (y-5)^2 = 100$

c. $(x+1)^2 + (y+5)^2 = 25$

d. $(x-4)^2 + (y-3)^2 = 25$

Center is the midpoint of AB: $M = \left(\frac{-3+5}{2}, \frac{2+8}{2}\right) = (1, 5)$. Radius squared is $(5-1)^2 + (8-5)^2 = 16 + 9 = 25$.

31 [Intermediate] • [MCQ] • Target: Chord Length from Distance to Center

In a circle of radius 13, a chord lies at a perpendicular distance of 5 from the center. What is the length of the chord?

a. 12

b. 18

c. 24 ✓

d. 26

Half the chord length is $\sqrt{13^2 - 5^2} = 12$, so the full chord length is $2(12) = 24$.

32 [Intermediate] • [MCQ] • Target: Trigonometric Ratios Across Quadrants

For a curriculum assessment problem, suppose if $\tan\theta = -\frac{8}{15}$ and $\sin\theta > 0$, what is the value of $\cos\theta$?

a. $-\frac{15}{17}$ ✓

b. $-\frac{8}{17}$

c. $\frac{15}{17}$

d. $\frac{8}{17}$

Since $\sin\theta > 0$ and $\tan\theta < 0$, θ is in Quadrant II where $\cos\theta < 0$. With opposite 8, adjacent 15, and hypotenuse 17, $\cos\theta = -\frac{15}{17}$.

33 [Intermediate] • [MCQ] • Target: Chord of Outer Circle Tangent to Inner Circle

Two concentric circles have radii 6 and 10. A chord of the larger circle is tangent to the smaller circle. What is the length of this chord?

a. 8

b. 12

c. 16 ✓

d. $2\sqrt{34}$

The radius of the smaller circle (6) is perpendicular to the chord at its midpoint, forming a right triangle with hypotenuse 10 (radius of the larger circle). Half-chord $= \sqrt{10^2 - 6^2} = 8$, so full chord length $= 16$.

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34 [Intermediate] • [MCQ] • Target: Angular Separation in Radians

What is the radian measure of the smaller angle between the hour hand and the minute hand of an analog clock at exactly 4:00?

a. $\frac{\pi}{3}$

b. $\frac{2\pi}{3}$ ✓

c. $\frac{3\pi}{4}$

d. $\frac{5\pi}{6}$

At 4:00, the hands are separated by $\frac{4}{12} = \frac{1}{3}$ of a full revolution, which is $\frac{1}{3}(2\pi) = \frac{2\pi}{3}$ radians (120°).

35 [Intermediate] • [MCQ] • Target: Rolling Wheel Linear Distance

A wheel of radius $0.4\,\text{m}$ rolls forward without slipping through 15 complete revolutions. How far does the center of the wheel travel?

a. $6\pi\,\text{m}$

b. $12\pi\,\text{m}$ ✓

c. $15\pi\,\text{m}$

d. $24\pi\,\text{m}$

Each revolution advances the wheel by one circumference $2\pi r = 0.8\pi\,\text{m}$. Over 15 revolutions, distance $= 15(0.8\pi) = 12\pi\,\text{m}$.

36 [Intermediate] • [MCQ] • Target: Circle Tangency to Coordinate Axes

If the circle $(x-4)^2 + (y-k)^2 = 25$ is tangent to the x-axis, what are the possible values of k?

a. ± 4

b. ± 5 ✓

c. ± 9

d. ± 25

A circle with center $(4, k)$ and radius $r = 5$ is tangent to the x-axis ($y = 0$) if and only if the vertical distance from its center to the x-axis equals its radius: $|k| = 5 \Rightarrow k = \pm 5$.

37 [Intermediate] • [MCQ] • Target: Simplifying Trigonometric Expressions

Which expression is equivalent to $\frac{1-\cos^2\theta}{\sin\theta\cos\theta}$ for all θ where the expression is defined?

a. $\sin\theta$

b. $\cos\theta$

c. $\tan\theta$ ✓

d. $\frac{1}{\tan\theta}$

Since $1 - \cos^2\theta = \sin^2\theta$, the expression simplifies to $\frac{\sin^2\theta}{\sin\theta\cos\theta} = \frac{\sin\theta}{\cos\theta} = \tan\theta$.

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Part II — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 **[Challenging] • [MCQ] • Target: Extremal Distance from a Point to a Circle • Cross-Lesson: Topic 7.01 (Distance Formula)**

What is the greatest distance from the point $P(7, 10)$ to any point on the circle $(x-1)^2 + (y-2)^2 = 9$?

a. 10 **b.** 13 ✓

c. 16 **d.** 19

The distance from $P(7, 10)$ to the center $C(1, 2)$ is $d = \sqrt{(7-1)^2 + (10-2)^2} = \sqrt{36+64} = 10$. Adding the radius $r = 3$ gives the maximum distance $d + r = 10 + 3 = 13$.

39 **[Challenging] • [MCQ] • Target: Intersecting Chords Theorem & Quadratics • Cross-Lesson: Topic 3.01 (Quadratic Equations)**

Two chords AB and CD intersect at point E inside a circle. If $AE = 4$, $EB = 9$, and $CE = x$, $ED = x + 5$, what is the length of chord CD?

a. 4 **b.** 9

c. 13 ✓ **d.** 15

By the Intersecting Chords Theorem, $AE \cdot EB = CE \cdot ED \Rightarrow 4(9) = x(x+5) \Rightarrow x^2 + 5x - 36 = 0 \Rightarrow (x+9)(x-4) = 0 \Rightarrow x = 4$. Thus $CD = CE + ED = 4 + 9 = 13$.

40 **[Challenging] • [MCQ] • Target: Coordinate Intercepts & Thales' Diameter Theorem • Cross-Lesson: Topic 1.01 (Intercepts) & Topic 7.01 (Right Triangles)**

In the xy-plane, the circle $x^2 + y^2 - 6x - 8y = 0$ intersects the x-axis at the origin $(0, 0)$ and at point A, and intersects the y-axis at the origin and at point B. What is the length of segment AB?

a. 5 **b.** 7

c. 10 ✓ **d.** 14

Since $\angle AOB = 90^\circ$ is an inscribed right angle, segment AB is a diameter of the circle! Completing the square gives $(x-3)^2 + (y-4)^2 = 25$, so $r = 5$ and diameter $AB = 10$ (indeed, $A(6, 0)$ and $B(0, 8)$ have distance $\sqrt{6^2 + 8^2} = 10$).

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41 [Challenging] • [MCQ] • Target: Pulley Belt Composite Perimeter • Cross-Lesson: Topic 8.01 (Composite Perimeter)

A belt wraps tightly around two congruent circular pulleys, each of radius $4\,\text{cm}$, whose centers are $14\,\text{cm}$ apart. What is the total length of the belt?

a. $14 + 8\pi$

b. $28 + 4\pi$

c. $28 + 8\pi$ ✓

d. $28 + 16\pi$

The belt consists of two straight tangent segments of length $14\,\text{cm}$ each ($28\,\text{cm}$ total) plus two 180° semicircular arcs of radius $4\,\text{cm}$ that together equal one full circumference $2\pi(4) = 8\pi\,\text{cm}$. Total length $= 28 + 8\pi\,\text{cm}$.

42 [Challenging] • [MCQ] • Target: Trigonometric Equations Quadratic in Form • Cross-Lesson: Topic 3.01 (Quadratic Factoring)

For how many values of x in the interval $0 \le x \le 2\pi$ does $2\sin^2(x) - \sin(x) - 1 = 0$ hold?

a. 1

b. 2

c. 3 ✓

d. 4

Factor as $(2\sin x + 1)(\sin x - 1) = 0$. On $[0, 2\pi]$, $\sin x = 1$ has 1 solution ($x = \frac{\pi}{2}$), and $\sin x = -\frac{1}{2}$ has 2 solutions ($x = \frac{7\pi}{6}, \frac{11\pi}{6}$). Thus there are $1 + 2 = 3$ solutions.

43 [Challenging] • [MCQ] • Target: Incircle of an Equilateral Triangle • Cross-Lesson: Topic 7.01 (Equilateral Triangles)

A circle is inscribed in an equilateral triangle of side length 12. What is the area of the circle?

a. 6π

b. 12π ✓

c. 18π

d. 36π

The altitude of the equilateral triangle is $h = 6\sqrt{3}$, and the inradius is one-third of the altitude: $r = \frac{1}{3}(6\sqrt{3}) = 2\sqrt{3}$. Therefore the circle's area is $\pi r^2 = \pi(2\sqrt{3})^2 = 12\pi$.

44 [Challenging] • [MCQ] • Target: Sector-to-Cone 3D Reconstruction • Cross-Lesson: Topic 8.01 (3D Cone Volume)

When a circular sector with central angle 216° and radius 10 is folded into a right circular cone, what is the volume of the cone?

a. 96π ✓

b. 128π

c. 144π

d. 288π

The cone's base radius satisfies $\frac{r}{10} = \frac{216^\circ}{360^\circ} = \frac{3}{5} \Rightarrow r = 6$. With slant height $\ell = 10$, the cone height is $h = \sqrt{10^2 - 6^2} = 8$, and its volume is $\frac{1}{3}\pi(6^2)(8) = 96\pi$.

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45 [Challenging] • [MCQ] • Target: Intersecting Circles & Rhombus Area • Cross-Lesson: Topic 8.01 (Rhombus Area)

If the circles $x^2 + y^2 = 9$ and $(x - 4)^2 + y^2 = 9$ intersect at points P and Q, what is the area of the rhombus $OPO'Q$ formed by the two centers $O(0, 0)$, $O'(4, 0)$ and the intersection points P, Q?

a. $4\sqrt{5}$ ✓

b. $6\sqrt{5}$

c. $8\sqrt{5}$

d. 12

All four sides of $OPO'Q$ are radii of length 3. The horizontal diagonal OO' has length 4, so half-diagonal is 2, and the vertical half-diagonal is $\sqrt{3^2 - 2^2} = \sqrt{5}$ (full vertical diagonal $2\sqrt{5}$). Rhombus area $= \frac{1}{2}(4)(2\sqrt{5}) = 4\sqrt{5}$.

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